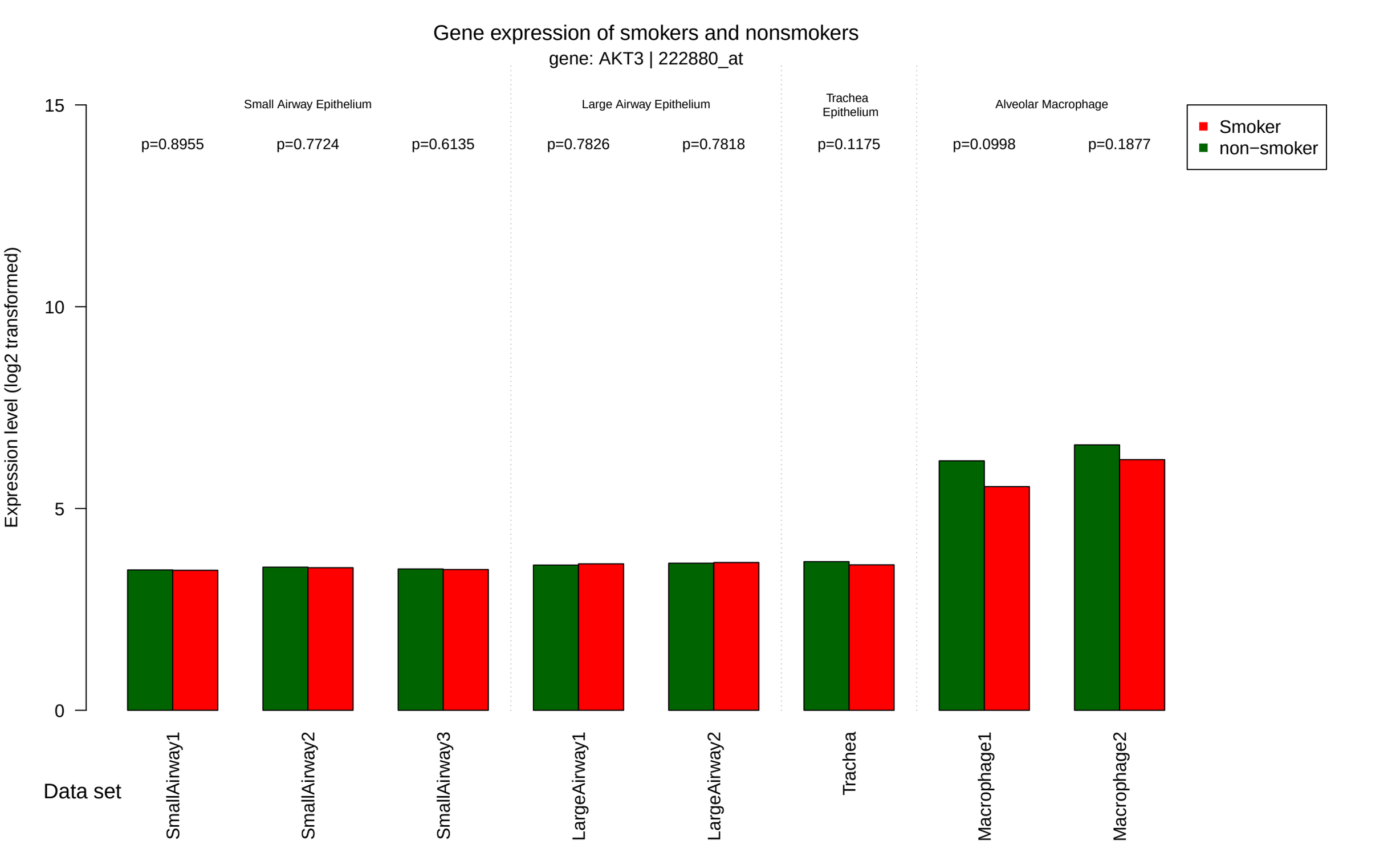
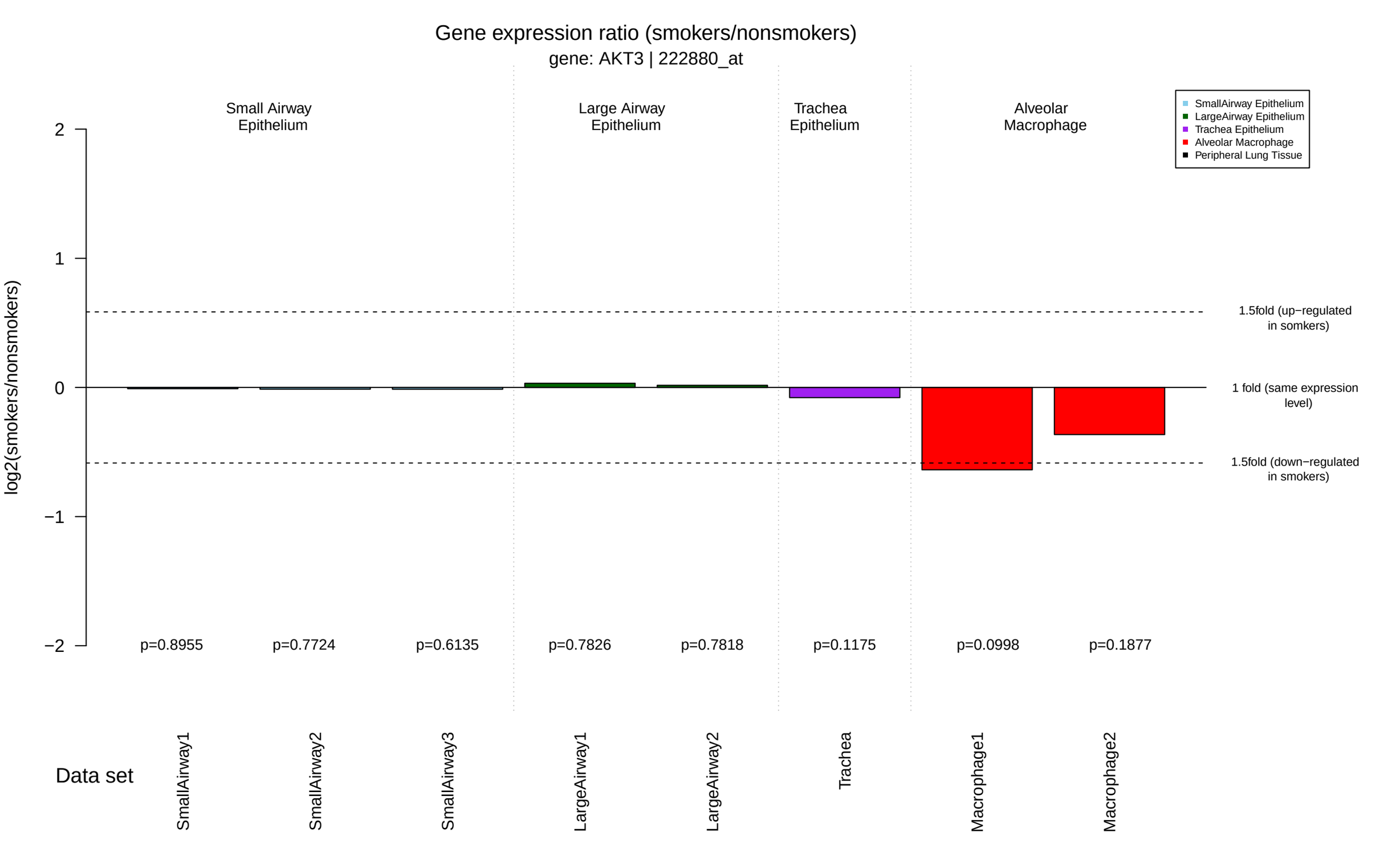
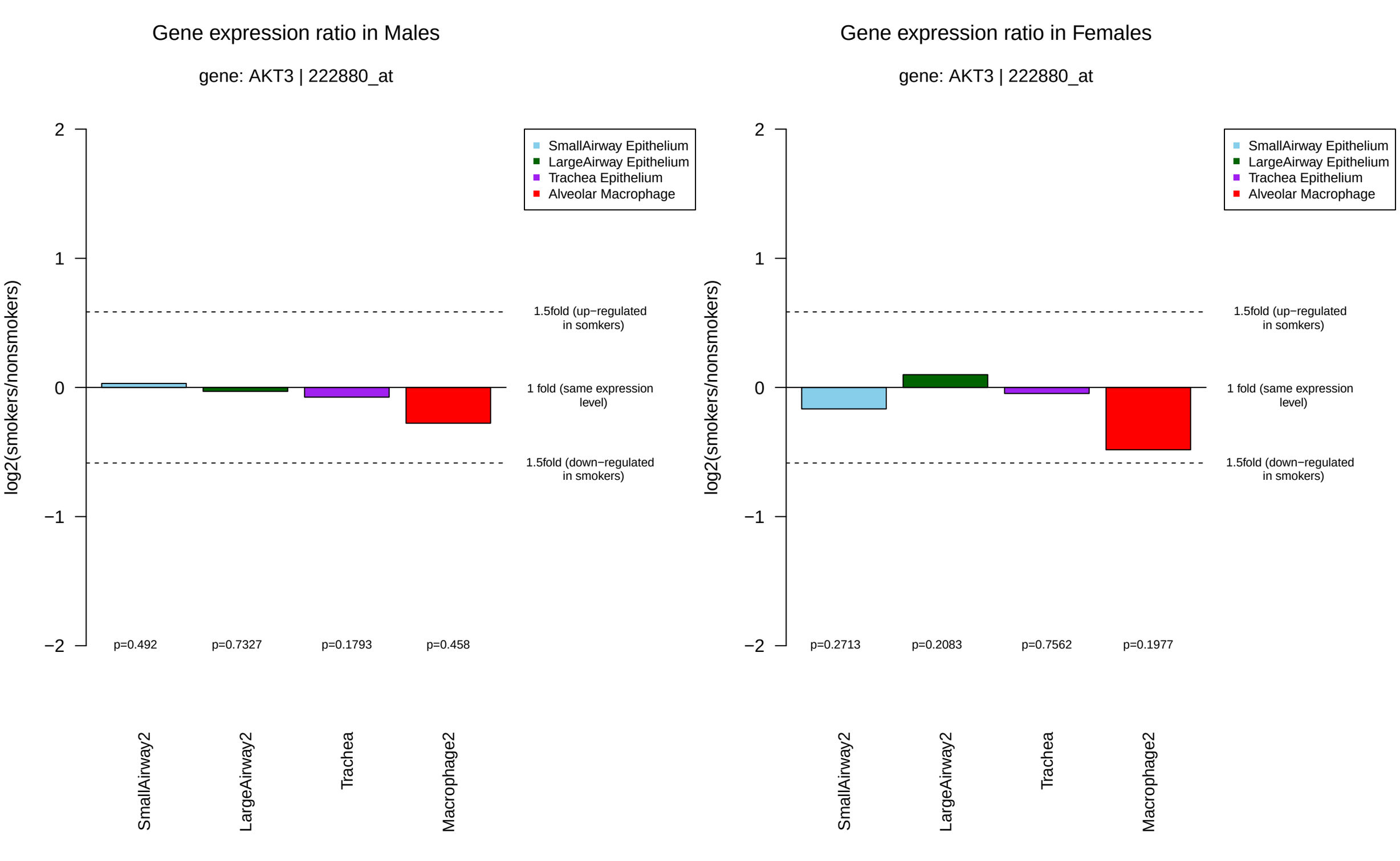
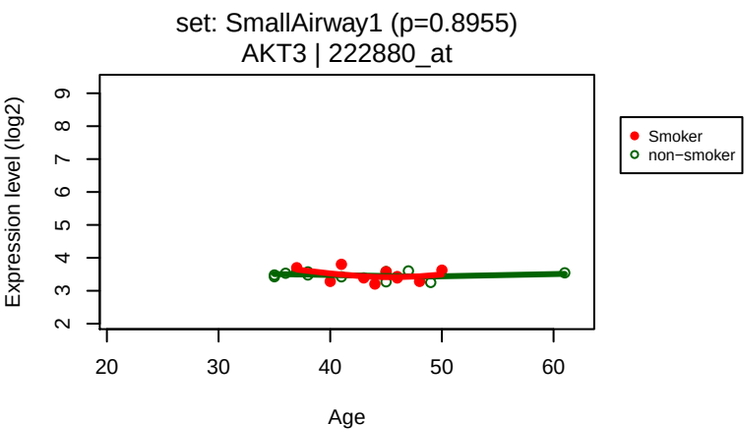
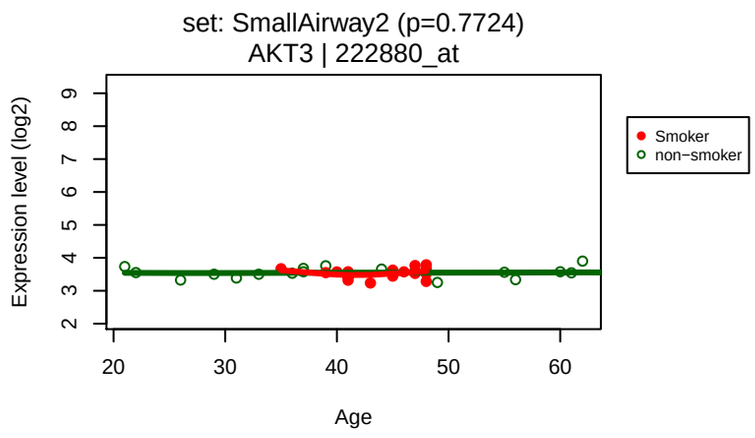
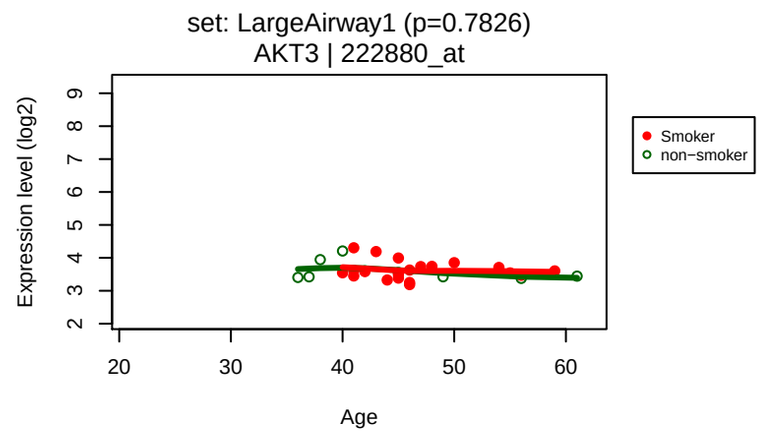

Gene expression of smokers and nonsmokers
gene: AKT3 | 222880_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8955
p=0.7724
p=0.6135
p=0.7826
p=0.7818
p=0.1175
p=0.0998
p=0.1877
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

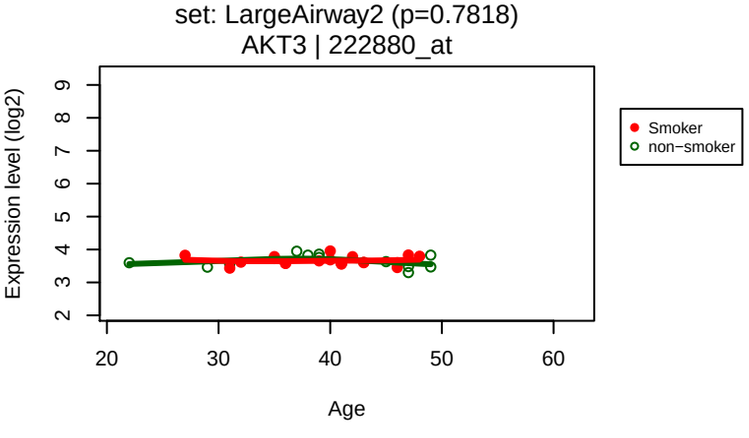

Gene expression ratio (smokers/nonsmokers)
gene: AKT3 | 222880_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8955
p=0.7724
p=0.6135
p=0.7826
p=0.7818
p=0.1175
p=0.0998
p=0.1877
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

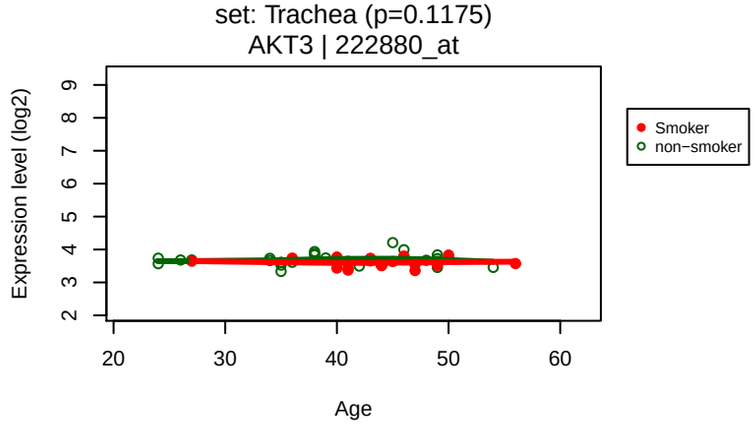

Gene expression ratio in Males
gene: AKT3 | 222880_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.492
p=0.7327
p=0.1793
p=0.458
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKT3 | 222880_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2713
p=0.2083
p=0.7562
p=0.1977
SmallAirway2
LargeAirway2
Trachea
Macrophage2

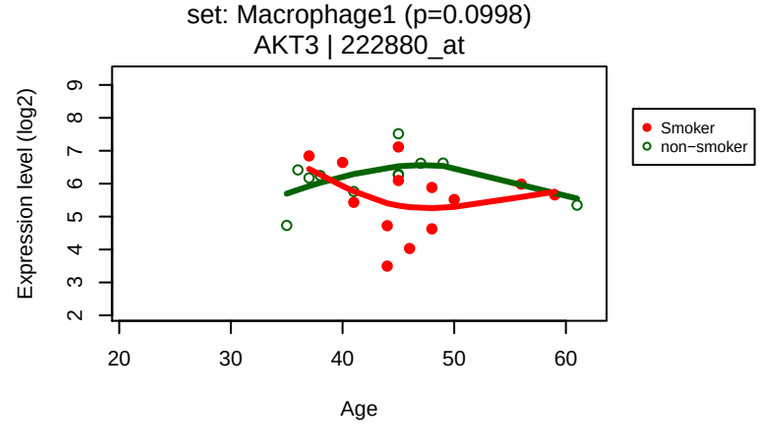

set: SmallAirway1 (p=0.8955)
AKT3 | 222880_at
Expression level (log2)
Age
Smoker
non−smoker

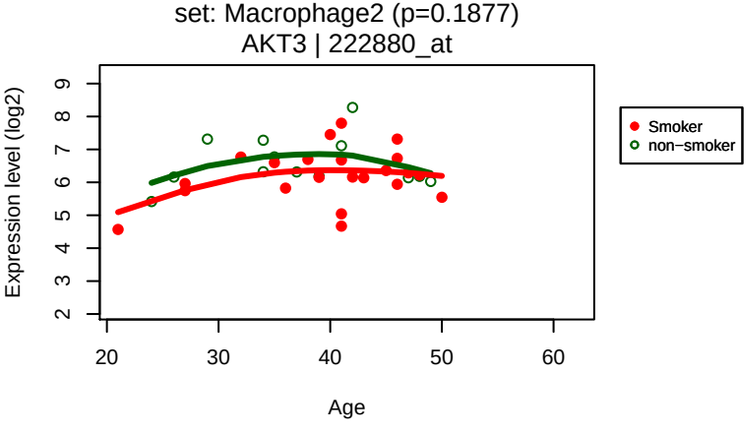

set: SmallAirway2 (p=0.7724)
AKT3 | 222880_at
Expression level (log2)
Age
Smoker
non−smoker

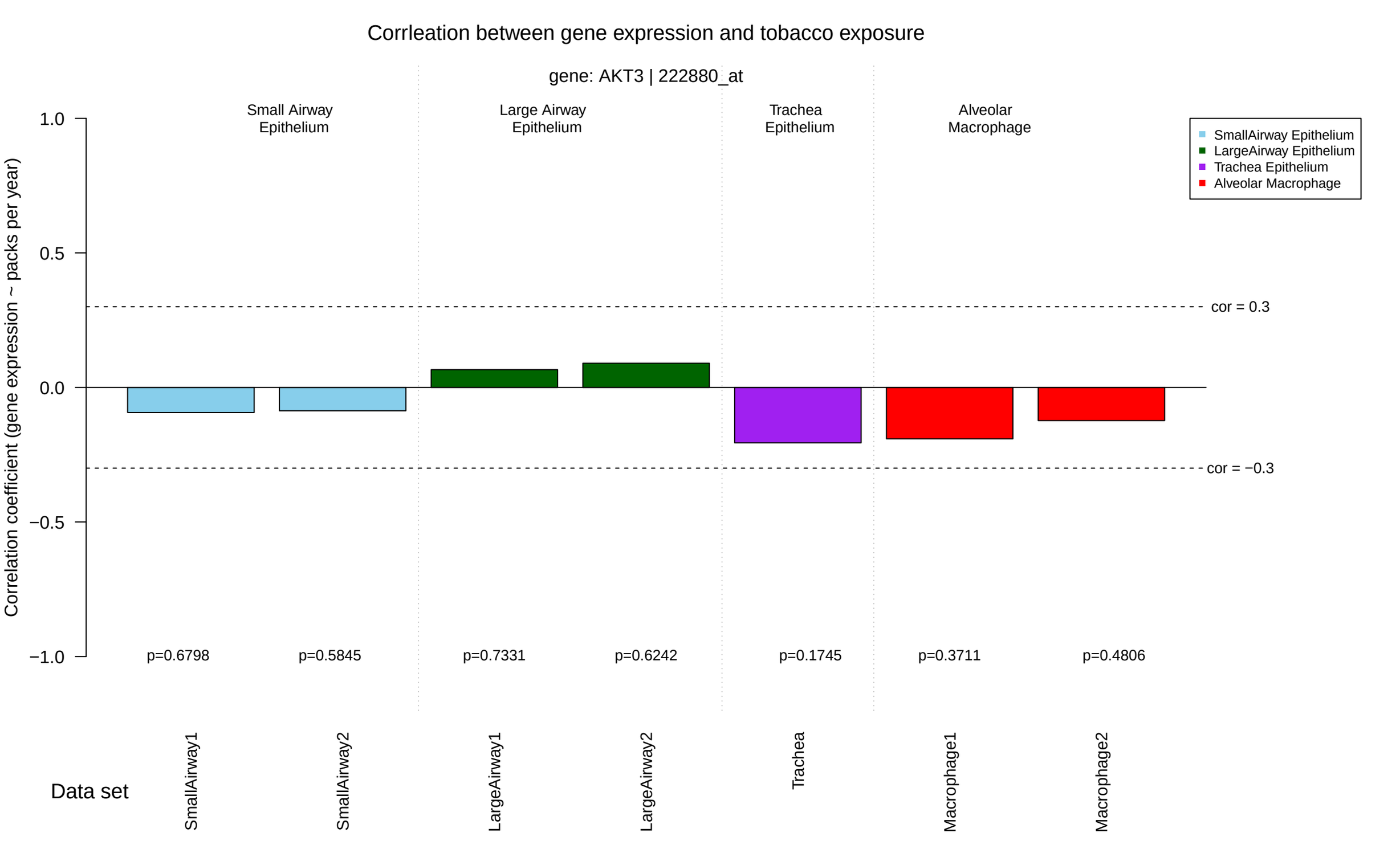

set: LargeAirway1 (p=0.7826)
AKT3 | 222880_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7818)
AKT3 | 222880_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1175)
AKT3 | 222880_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0998)
AKT3 | 222880_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1877)
AKT3 | 222880_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKT3 | 222880_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6798
p=0.5845
p=0.7331
p=0.6242
p=0.1745
p=0.3711
p=0.4806
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2